

An illustration within a white rounded rectangle showing three stylized figures climbing a bar chart. The chart has three bars of increasing height: a small blue bar, a medium orange bar, and a tall blue bar. A man in a blue shirt and orange pants is on the ground, reaching up to help a woman in a blue shirt and dark pants who is standing on the orange bar. Another man in an orange shirt and dark pants is standing on the top of the tallest blue bar, reaching down to help the woman. To the left of the bars is a potted plant with large green leaves. The background of the illustration is a light orange semi-circle.

APIS, DATA MERGING, AND TRANSFORMATION

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LECTURE OVERVIEW

- Using APIs for data retrieval
- Merging clinical and omics data
- Data normalization techniques



INTRODUCTION

In this session, we'll explore how to retrieve data using APIs, merge diverse data types like clinical and omics data, and apply normalization techniques. These skills are essential for data-driven biomedical research. We will focus on preparing raw data for downstream tasks like analysis or machine learning through real-world bioinformatics workflows.



WHY INTEGRATING DATA MATTERS

- Modern biomedical research often involves multiple types of data collected from the same subjects—clinical attributes like age or disease stage and omics data such as gene expression profiles. These data types are often stored separately, in different formats, and at different levels of complexity. Merging and normalizing such data is crucial to gain meaningful insights and reduce analytical bias.



WHAT IS AN API?

- An Application Programming Interface (API) allows software to communicate and exchange data. In the context of biomedical research, APIs allow you to programmatically access large public databases, such as those containing genomic or clinical datasets. Rather than downloading data manually, APIs allow you to query exactly what you need using simple code.



BENEFITS OF APIS

- Automation of Data Retrieval
 - It helps to programmatically access data, eliminating the need for manual downloads. You can write a script once and reuse it anytime to fetch the most up-to-date data.
- Precise and Filtered Access
 - APIs let you query exactly what you need, for example, gene expression data for a specific cancer type or clinical metadata for a subset of patients—using parameters like disease name, gene symbol, or file type. This prevents downloading massive files with unnecessary content
- Integration into Pipelines
 - Because APIs return structured formats like JSON or CSV, they can be easily integrated into **bioinformatics pipelines** or machine learning workflows. This ensures a seamless transition from raw data retrieval to downstream processing, analysis, and visualization.



COMMON APIS IN BIOMEDICAL DATA

Major Databases	RESTful APIs
The Cancer Genome Atlas (TCGA) dataset	GDC API to query multi-omics and clinical data
cBioPortal	cBioPortal API to query cancer genomics and clinical data
Ensembl	Ensembl REST API- This offers genetic and disease-related annotations
OMIM	OMIM API - Same as Ensembl

A **RESTful API** (REpresentational State Transfer) is a type of **web API** that uses standard HTTP methods like GET, POST, PUT, and DELETE to access and manipulate resources.

It is the most commonly used API style on the web today.



EXAMPLE- GDC API IN PYTHON

- Using Python's requests library, you can easily query the GDC API to retrieve data like file names, gene expression matrices, or clinical metadata. You send your query as parameters in a URL and receive structured JSON in return. This enables fast, automated access to large datasets, which can then be parsed and analyzed within your own code environment.
- For more information: https://docs.gdc.cancer.gov/API/Users_Guide/Getting_Started/

arduino

 Copy  Edit

```
https://api.example.com/data?query=GENE123
```


COMMON REST CODES

CODE

- 200
- 400
- 403
- 500

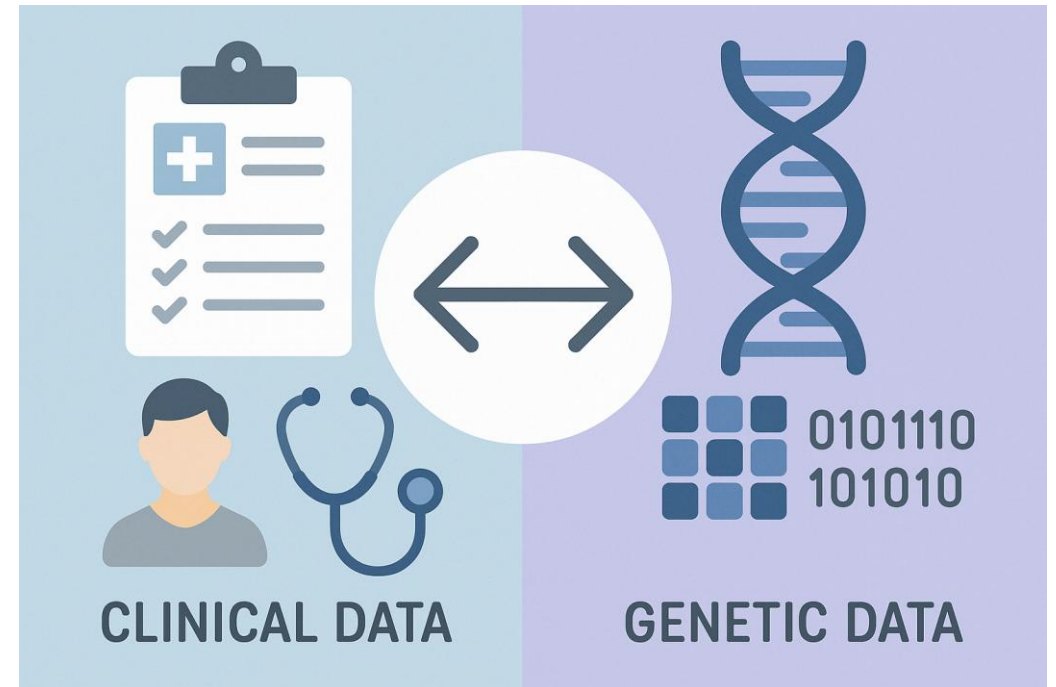
NAME AND MEANING

- OK | THE REQUEST WORKED!
- BAD REQUEST | THE SERVER DID NOT UNDERSTAND YOUR REQUEST
- FORBIDDEN | YOU ARE LOGGED IN BUT NOT ALLOWED TO DO THIS
- INTERNAL SERVER ERROR | THE SERVER CRASHED (IT'S NOT YOUR FAULT)



DATA MERGING – WHY ?

- To gain comprehensive insights, it's essential to connect clinical characteristics—such as age, gender, or cancer stage—with biological data like gene expression. Merging these datasets enables integrated analyses, such as identifying biomarkers associated with disease progression. Without merging, any insights drawn from isolated data types may be incomplete or misleading.
- **Understanding Data Formats:**
 - Clinical data is typically structured and low-dimensional, often stored in spreadsheets with one row per patient. Omics data, in contrast, is high-dimensional and can contain thousands of features like genes or proteins. Effective integration of these datasets depends on shared identifiers such as sample ID or patient ID, and requires cleaning and aligning data formats





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• PRACTICAL SESSION



MEET THE INSTRUCTORS

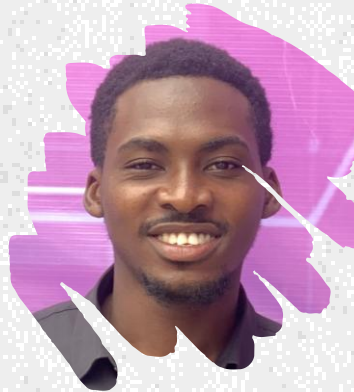


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THANK YOU

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